Collaborative Study Material and Online Quiz Platform — Component Diagram

One view of the whole system: every functional requirement (FR-01 to FR-14) and non-functional requirement (NFR-01 to NFR-07), organised in three layers. Tags on each component name the requirements it realises.

Layer 1 · User Interface

screens used by Student · Authorized User · Admin/Moderator (NFR-03 usability)

- «component» **Register / Log in** — #/register · #/login — FR-01
- «component» **Profile & Leaderboard** — contributions · reputation — FR-01 · FR-13
- «component» **Channels** — find · create · join · members — FR-02 · FR-03
- «component» **Study Materials** — search · upload · preview; rate · review · report — FR-04
- «component» **Question Bank** — search · add · rate · report — FR-05
- «component» **Quizzes** — create · publish · attempt; result · history — FR-05 · FR-06
- «component» **Ask AI & Review Drafts** — explain · topic quiz; approve generated questions — FR-07 · FR-08 · FR-09
- «component» **Chat** — channel threads · DMs; share material / quiz — FR-10 · FR-11
- «component» **Performance** — trends · weak topics; recommendations — FR-12
- «component» **Moderation Queue** — reports · moderator actions — FR-13
- «component» **Admin Console** — users · channels · content — FR-14
- «component» **Notifications** — inbox · preferences — FR-14

user actions (click · type)

Layer 2 · Frontend

client-side application in the browser — plain HTML/CSS/JS, no build step, served by the backend at /app/

- «component» **Hash Router** — app.js · one function per page — NFR-03
- «component» **Page Modules** — app.js · study.js; StudyHubPages registry — NFR-03 · NFR-07

uses

- «component» **Safe DOM Builder** — h() + textContent; no innerHTML · CSP self — NFR-02
- «component» **SVG Chart Renderer** — performance trends; weak-topic charts — NFR-03 · FR-12
- «component» **Permission Cache** — /accounts/me/permissions; UI hints only — NFR-02
- «component» **Quiz Timer & Autosave** — server-owned clock; saves each answer via API — NFR-04 · FR-06
- «component» **API Client** — api.js · get/post/upload/blob; ApiError · token expiry — NFR-01 · NFR-07
- «component» **Session Store** — bearer token (per tab); read by API Client — NFR-02

serves HTML / CSS / JS to the browser

REST /api/v1 · HTTPS + JSON + Bearer token

Layer 3 · Backend

FastAPI server · modular monolith · stateless and horizontally scalable (NFR-05)

API layer

- «component» **Static File Server** — serves /app/ with; CSP security headers — NFR-02
- «component» **Auth Guards** — current_user · require_capability; on every protected route — NFR-02 · FR-01
- «component» **REST Routers /api/v1** — pydantic validation · errors; described by OpenAPI v1 — NFR-01 · NFR-07
- «component» **OpenAPI Contract v1** — docs/api/openapi-v1.json; checked in CI — NFR-07

resolve user · capability

service calls

Feature modules

each owns its tables and exposes one public service.py · boundaries enforced by test_module_boundaries.py

- «component» **profiles** — contribution history — FR-01
- «component» **channels** — normal / authorized; approval · membership — FR-02 · FR-03
- «component» **materials** — files · search; ratings · reviews — FR-04
- «component» **accounts** — users · roles; login · tokens — FR-01
- «component» **questions** — question bank; LLM drafts — FR-05 · FR-07
- «component» **quizzes** — attempts · deterministic scoring; reads questions via service — FR-05 · FR-06
- «component» **llm** — generate · adapt; evaluate · explain — FR-07 · FR-08 · FR-09
- «component» **notifications** — alerts · preferences — FR-14
- «component» **chat** — threads · DMs · blocks; shared resource links — FR-10 · FR-11
- «component» **moderation (+ queue)** — reports · moderator actions — FR-13
- «component» **reputation** — points · badges; leaderboard — FR-13
- «component» **admin** — manage users, channels, content — FR-14
- «component» **analytics** — trends · weak topics; recommendations — FR-12

membership

llm → questions

attempts

hashing · tokens

own tables via ORM

material files

Infrastructure & data

- «component» **Security Core** — scrypt hashing · HS256; tokens · env secrets — NFR-02
- «component» **PostgreSQL 16** — transactions · constraints; indexes — NFR-04 · NFR-05
- «component» **ORM & Migrations** — SQLAlchemy 2 · Alembic (reversible) — NFR-04 · NFR-07
- «component» **File Store** — material files and pictures in PostgreSQL tables — NFR-04 · SEC-08
- «component» **LLM Provider Adapter** — timeouts · fallback; fake provider for tests — NFR-06

SQL

HTTPS

- «external» **LLM Service** — third-party API — external

Non-functional requirements → components

NFR-01 Performance
- API Client · REST Routers
- PostgreSQL indexes
- in-process deterministic scoring

NFR-02 Security
- Auth Guards · Security Core
- Session Store · Safe DOM Builder
- Static Server CSP · Permission Cache

NFR-03 Usability
- all Layer 1 screens · Hash Router
- Page Modules · SVG Chart Renderer

NFR-04 Reliability
- Quiz Timer & Autosave · PostgreSQL
- transactions · File Store
- reversible migrations

NFR-05 Scalability
- stateless backend (token auth)
- PostgreSQL · modules can be
- extracted into services

NFR-06 LLM reliability
- llm review gate (drafts)
- LLM Provider Adapter
- deterministic quiz scoring

NFR-07 Maintainability
- feature modules + service.py
- OpenAPI Contract v1
- ORM & Migrations · Page Modules

Legend

- User interface screen
- Frontend (browser) component
- Backend API component
- Backend feature module
- Infrastructure / data store
- External system
- HTTPS request (REST interface)
- uses / calls (dependency)
- secondary use / serves
- describes (contract)
- grouping inside a layer

«component» box with ⧉ icon = UML component

FR-xx / NFR-xx = requirements it realises